

WDC-ASM-003 WRCG/CST Projection Verification Record v0.1

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Self-Evo WDC v0.1.2

Source file:

controls/gate_records/WDC_ASM_003_WRCG_CST_PROJECTION_VERIFICATION_RECORD_v0_1.m
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WDC-ASM-003 WRCG/CST Projection Verification Record v0.1

Gate-execution verification record for WDC-ASM-003 / SEARG-WDC-0Q-005

Status: Package-control gate-execution candidate / verification record

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Record ID: WDC_ASM_003_WRCG_CST_PROJECTION_VERIFICATION_RECORD_v0_1

Short name: WDC-ASM-003-PROJECTION-VERIFY v0.1

Gate under verification: WDC-ASM-003 — verify 07 WRCG/CST projection to 03 packet-facing enums

Open issue source: SEARG-WDC-0Q-005 / SE-0I-WDC-003

Decision posture: verification candidate only; not gate closure, not package assembly, not release

Primary source artifact: 07_Self_Evo_Anti_Autarky_and_Resource_Gate_WDC_Delta_v0_1_2_v0_1_1.md

Primary source SHA-256: 804a8e341e30a2a86fa43429618f24e62458366e236c7b6ef8c12762d94da7af

Packet-schema target: 03_Self_Evo_Proposal_Packet_Schema_WDC_Delta_v0_1_2_v0_1_2.md

Packet-schema SHA-256: 5a7e6af6bc83d3b7ccde875eae1a0e928600cf509b494bd773c43c29537264f3

> This record verifies the projection model. It does not close WDC-ASM-003 by itself. Closure requires a separate review and a c/a gate decision, following the pattern used for WDC-ASM-002.

0. Executive summary

WDC-ASM-003 exists because 07 has richer resource-gate sidecar states than the packet-facing schema in 03.

The question is narrow:

Can every WRCG gate-record result and every CST gate-record outcome in 07 project into the accepted 03 packet enums without leaving unprojectable values or hiding material WDC risk?

Answer of this verification record:

YES — as a gate-execution candidate.

The projection is:

total: yes
unprojectable values found: none
packet enum compatibility: yes
diagnostic preservation: yes, through sidecar state + canonical annotation + checker route
fail-closed behavior: yes
textual update required to 03: no
textual update required to 07: no

Important precision:

The packet field alone is intentionally compact and therefore not semantically injective.
The package-level projection is lossless only because 07 retains the richer gate-record sidecar and binds compressed packet states to canonical annotations and checker routes.

Therefore this record recommends:

WDC-ASM-003 candidate: SOUND
recommended next artifact: WDC_ASM_003_WRCG_CST_PROJECTION_VERIFICATION_REVIEW_RECORD_v0_1.md

1. Boundary

This file is a package-control verification record.

It does not:

```
close WDC-ASM-003;
assemble the package;
release v0.1.2;
change 03, 07, or 04;
claim conformance;
claim deployment readiness;
certify witness independence;
authorize live self-evolution;
write memory;
replace c/a decision authority.
```

It verifies whether the existing 07 projection crosswalk is compatible with the accepted 03 packet schema and the canonical 04 checker vocabulary.

2. Corpus bridge set

2.1 Explicit bridge: $c = a + b$

In $c = a + b$, WDC-ASM-003 belongs to b.

It is not the human anchor a, and it is not the living decision contour c. It is a documentary control surface that prevents one layer's richer operational language from silently becoming another layer's invalid schema value.

The purpose is to keep b honest:

```
07 may speak in resource-gate detail.
03 must receive packet-safe values.
04 must receive canonical checker outcomes.
No layer may launder an unresolved witness-resource risk into a clean packet pass.
```

2.2 Quiet bridge I: information theory

Projection is compression.

A WRCG sidecar result such as `pass_with_gates` carries more information than the packet value `pass`. If that extra information is discarded, the channel becomes lossy in the dangerous sense: a gated pass and a clean pass become indistinguishable.

This record therefore checks not only the compressed packet value, but also the companion annotation, evidence reference, and checker route that preserve the lost bits.

2.3 Quiet bridge II: cybernetics

A control loop can use a reduced signal only if the omitted state remains available somewhere else in the loop.

Here the packet enum is the reduced signal. The sidecar, WDC annotation, and checker route carry the feedback state. If `warning`, `hold`, or `unverified` were projected only to a harmless packet value with no route, the regulator would lose negative feedback and the package would drift toward false readiness.

2.4 Quiet bridge III: physiology

A blood test summary can say "within range," but the clinician still needs the underlying marker if the patient is under stress.

The packet result is the summary marker. The sidecar and annotation are the underlying labs. The patient is not discharged because one value fits in a compact field.

2.5 Earth paragraph

On a construction site, a junction box label may use a short code, but the as-built folder must still map that code to breaker, circuit, load, room, inspection note, and commissioning status. If the label says OK while the commissioning sheet says "temporary bypass," someone eventually energizes the wrong line. This verification is the as-built mapping for WRCG and CST: short packet values are allowed only when the longer safety meaning is still recoverable.

3. Source basis

```
| Role | File | Lines | SHA-256 | Use in this verification |
|---|---|---|---|---|
| `07` resource-gate WDC delta | `07_Self_Evo_Anti_Autarky_and_Resource_Gate_WDC_Delta_v0_1_2_v0_1_1.md` | 1309 | `804a8e341e30a2a86fa43429618f24e62458366e236c7b6ef8c12762d94da7af` | WRCG/CST source vocabulary and projection crosswalk |
| `03` packet schema WDC delta | `03_Self_Evo_Proposal_Packet_Schema_WDC_Delta_v0_1_2_v0_1_2.md` | 1003 | `5a7e6af6bc83d3b7ccde875eae1a0e928600cf509b494bd773c43c29537264f3` | Packet-facing accepted enums |
| `04` canonical checker WDC delta | `04_Self_Evo_Local_Checker_Rules_WDC_Delta_v0_1_2_v0_1_1.md` | 1154 | `fcb7826ea432473b5677faf7b46cc5a1c47400c02fcec24429794b15064af40` | Canonical WDC checker result / annotation route |
| `08` fixture WDC delta | `08_Self_Evo_Conformance_Fixture_Pack_WDC_Delta_v0_1_2_v0_1_2.md` | 1419 | `4aa5b85e0a0af4c06fc56f30ba355ec6c378133fe3d70caee352fb93b3944389` | Fixture-output compatibility boundary |
| patch index | `SELF_EVO_v0_1_2_PATCH_INDEX_v0_1_3.md` | 855 | `af01b1b443a87c111ab7dab2e001d01e8bffc0710362b89da7781238447ea28b` | Harmonized package index after WDC-ASM-002 |
| package manifest | `SELF_EVO_WDC_v0_1_2_PACKAGE_ASSEMBLY_MANIFEST_v0_1_3.md` | 688 | `aa5fb978000d96df74c03dc735bcbd1c8170b9d34644e85b8297fd114759f917` | Current manifest: WDC-ASM-002 closed, WDC-ASM-003 pending |
| WDC-ASM-002 closure | `WDC_ASM_002_GATE_CLOSURE_RECORD_v0_1.md` | 608 | `6c4c92516162043a562983fe128be13a0cf965a3d17659d363d1cb0142d96107` | Prior gate closure precedent |
```

4. Verification question

SEARG-WDC-0Q-005 asks package assembly to verify:

```
WRCG gate-record result -> witness_resource_change_gate.result
CST gate-record outcome -> challenge_survivability_test.result
```

The verification criteria used here are:

1. the 03 packet enums are correctly identified;
2. every 07 WRCG gate-record result has a packet projection;
3. every 07 CST gate-record outcome has a packet projection;
4. richer states compressed into packet values retain diagnostics through sidecar state, canonical annotation, and checker route;
5. unknown or unprojectable states fail closed or hold/human-gate; they do not become clean pass;
6. PASS_WITH_LIMITS is not used as a checker, WRCG, CST, or fixture output;
7. no textual change to 03 or 07 is required for the projection to be valid.

5. Accepted 03 packet enums

5.1 witness_resource_change_gate.result

03 accepts:

```
{pass, fail, human_gate, not_run}
```

This is the only packet-facing enum for WRCG.

5.2 challenge_survivability_test.result

03 accepts:

```
{pass, fail, not_run, not_applicable}
```

This is the only packet-facing enum for CST.

5.3 Consequence

Any value outside these sets must either:

```
project into one of these values with preserved diagnostic sidecar context;
or fail closed as unprojectable.
```

6. WRCG projection verification

6.1 Source and target

Source field in 07:

```
WRCG gate-record result
```

Packet target field in 03:

```
witness_resource_change_gate.result
```

Accepted target enum:

```
{pass, fail, human_gate, not_run}
```

6.2 Verified WRCG projection table

```
| `07` WRCG gate-record result | Packet projection to `03` | Required diagnostic preservation | Checker
route | Verification |
|---|---|---|---|
| `pass` | `pass` | none beyond ordinary evidence refs | `PASS` | valid |
| `pass_with_gates` | `pass` | `wdc.clean_with_gates` + required gate refs | `PASS_WITH_GATES` | valid,
compressed but preserved |
| `human_gate` | `human_gate` | human-anchor review ref | `HUMAN_GATE` | valid |
| `warning` | `pass` | `wdc.challenge_cost_mitigation_missing` report-visible annotation | `WARNING` |
valid, compressed but preserved |
| `fail` | `fail` | failure reason / WRCG or WRF evidence | `FAIL` | valid |
| `hold` | `not_run` | `wdc.relevance_unknown` annotation | `HOLD` | valid, compressed but preserved |
| `not_run` | `not_run` | trigger/materiality context | `HOLD` or no WDC route depending trigger state |
valid |
```

6.3 WRCG totality result

```
wrcg_projection_verification:
  source_values_checked:
    - pass
    - pass_with_gates
    - human_gate
    - warning
    - fail
    - hold
    - not_run
  target_enum: [pass, fail, human_gate, not_run]
  unprojectable_source_values: []
  packet_projection_total: true
  semantic_preservation_method:
    - sidecar_result
    - canonical_wdc_annotation
    - checker_route
    - evidence_ref_or_reviewer_ref
  packet_projection_alone_is_injective: false
  package_control_projection_is_lossless: true
```

6.4 WRCG fail-closed rule

If a WRCG gate-record result cannot be projected into the 03 packet enum, 07 requires:

```
canonical_result: FAIL
annotation: wdc.wrcg_missing_or_failed
reason: unprojectable_wrcg_result
```

This is acceptable because unprojectable state becomes a visible fail route, not a silent pass.

7. CST projection verification

7.1 Source and target

Source field in 07:

```
CST gate-record outcome
```

Packet target field in 03:

```
challenge_survivability_test.result
```

Accepted target enum:

```
{pass, fail, not_run, not_applicable}
```

7.2 Verified CST projection table

```
| `07` CST gate-record outcome | Packet projection to `03` | Required diagnostic preservation | Checker
route | Verification |
|---|---|---|---|---|
| `verified` | `pass` | CST evidence ref | may support `PASS_WITH_GATES` | valid |
| `unverified` | `not_run` | `wdc.challenge_survivability_unverified` annotation | `HUMAN_GATE` via
`WDC-LC-005` | valid, compressed but preserved |
| `failed` | `fail` | CST failure evidence | `FAIL` via WRF/WRCG or stricter route | valid |
| `not_applicable` | `not_applicable` | no witness-resource impact context | ordinary SEARG route |
valid |
| `unknown` | `not_run` | `wdc.relevance_unknown` annotation | `HOLD` | valid, compressed but preserved
|
| `not_run` | `not_run` | materiality / trigger context | `HOLD` or `HUMAN_GATE` depending materiality |
valid |
```

7.3 CST totality result

```
cst_projection_verification:
  source_values_checked:
    - verified
    - unverified
    - failed
    - not_applicable
    - unknown
    - not_run
  target_enum: [pass, fail, not_run, not_applicable]
  unprojectable_source_values: []
  packet_projection_total: true
  semantic_preservation_method:
    - sidecar_outcome
    - canonical_wdc_annotation
    - checker_route
    - evidence_ref
    - materiality_context
  packet_projection_alone_is_injective: false
  package_control_projection_is_lossless: true
```

7.4 CST fail-safe rule

If CST is material and the only packet projection is not_run, the checker must not treat that as clean pass.

It must route through:

```
WDC-LC-005
annotation: wdc.challenge_survivability_unverified
```

or a stricter failure route if the facts prove the witness cannot survive challenge.

8. Cross-layer compatibility matrix

```
| Case | `07` sidecar value | `03` packet value | `04` canonical annotation / route | Risk of hidden
pass? | Verdict |
|---|---|---|---|---|
| clean WRCG | `pass` | `pass` | `PASS` / clean route | no | compatible |
| gated WRCG | `pass_with_gates` | `pass` | `wdc.clean_with_gates` / `PASS_WITH_GATES` | no, if
sidecar+annotation retained | compatible |
| human WRCG | `human_gate` | `human_gate` | `HUMAN_GATE` | no | compatible |
| challenge-cost warning | `warning` | `pass` | `wdc.challenge_cost_mitigation_missing` / `WARNING` |
no, if annotation retained | compatible |
| WRCG failure | `fail` | `fail` | `wdc.wrcg_missing_or_failed` or stricter | no | compatible |
| unknown relevance | `hold` | `not_run` | `wdc.relevance_unknown` / `HOLD` | no | compatible |
| WRCG absent | `not_run` | `not_run` | `HOLD`, `FAIL`, or no WDC route depending trigger | no, if
materiality checked | compatible |
| CST verified | `verified` | `pass` | supports gated pass | no | compatible |
| CST unverified | `unverified` | `not_run` | `wdc.challenge_survivability_unverified` / `HUMAN_GATE` |
no | compatible |
| CST failed | `failed` | `fail` | `FAIL` | no | compatible |
| CST unknown | `unknown` | `not_run` | `wdc.relevance_unknown` / `HOLD` | no | compatible |
| CST absent | `not_run` | `not_run` | `HOLD` or `HUMAN_GATE` depending materiality | no, if materiality
checked | compatible |
```

9. PASS_WITH_LIMITS discipline

PASS_WITH_LIMITS must remain a review-record disposition only.

For WDC-ASM-003, it is invalid as:

```
WRCG gate-record result;
CST gate-record outcome;
03 packet enum;
04 checker result;
08 fixture expected output;
manifest gate state.
```

Verification result:

```
pass_with_limits_usage:
  allowed_contexts:
    - review history
    - review record disposition
    - prohibition statement
  forbidden_contexts:
    - checker output
    - fixture output
    - WRCG result
    - CST outcome
    - packet enum
  status: compatible
```

10. Manifest impact

When this gate is reviewed and closed by c/a, the manifest may update:

```
WDC-ASM-003: CLOSED
projection_verified: true
```

It must not update:

```
package_assembled: true
public_release_authorized: true
conformance_certified: true
deployment_authorized: true
witness_independence_certified: true
```

Those remain false until their own gates close.

The manifest currently keeps WDC-ASM-003 pending. This record does not change that state by itself.

11. Machine-readable verification object

```
wdc_asm_003_projection_verification_record:
  schema_version: &quot;wdc-asm-003-projection-verification-record-0.1&quot;;
  record_id: &quot;WDC_ASM_003_WRCG_CST_PROJECTION_VERIFICATION_RECORD_v0_1&quot;;
  created_at: &quot;2026-06-27T18:17:24Z&quot;;
  record_class: &quot;package_control_gate_execution_candidate / projection_verification /
    witness_compatible&quot;;
  gate_under_verification: &quot;WDC-ASM-003&quot;;
  source_issue: &quot;SEARG-WDC-0Q-005 / SE-OI-WDC-003&quot;;

reviewed_source_artifacts:
  searg_07_wdc_delta:
    path: &quot;07_Self_Evo_Anti_Autarky_and_Resource_Gate_WDC_Delta_v0_1_2_v0_1_1.md&quot;;
    sha256: &quot;804a8e341e30a2a86fa43429618f24e62458366e236c7b6ef8c12762d94da7af&quot;;
    line_count: 1309
  packet_schema_03_wdc_delta:
    path: &quot;03_Self_Evo_Proposal_Packet_Schema_WDC_Delta_v0_1_2_v0_1_2.md&quot;;
    sha256: &quot;5a7e6af6bc83d3b7ccde875eae1a0e928600cf509b494bd773c43c29537264f3&quot;;
    line_count: 1003
  canonical_checker_04_wdc_delta:
    path: &quot;04_Self_Evo_Local_Checker_Rules_WDC_Delta_v0_1_2_v0_1_1.md&quot;;
    sha256: &quot;fcb7826ea432473b5677faf7b46cc5a1c47400c02fcec24429794b15064af40&quot;;
    line_count: 1154
  manifest:
    path: &quot;SELF_EVO_WDC_v0_1_2_PACKAGE_ASSEMBLY_MANIFEST_v0_1_3.md&quot;;
    sha256: &quot;aa5fb978000d96df74c03dc735bcbd1c8170b9d34644e85b8297fd114759f917&quot;;
    line_count: 688

packet_enums:
  witness_resource_change_gate.result:
    - pass
    - fail
    - human_gate
    - not_run
  challenge_survivability_test.result:
    - pass
    - fail
    - not_run
    - not_applicable

wrcg_projection:
  pass:
    packet: pass
    checker_route: PASS
  pass_with_gates:
    packet: pass
    annotation: wdc.clean_with_gates
    checker_route: PASS_WITH_GATES
  human_gate:
    packet: human_gate
    checker_route: HUMAN_GATE
  warning:
    packet: pass
    annotation: wdc.challenge_cost_mitigation_missing
    checker_route: WARNING
  fail:
    packet: fail
    checker_route: FAIL
  hold:
    packet: not_run
    annotation: wdc.relevance_unknown
    checker_route: HOLD
  not_run:
    packet: not_run
    checker_route: HOLD_or_no_WDC_routeDepending_on_trigger_state

cst_projection:
  verified:
    packet: pass
    checker_route: may_support_PASS_WITH_GATES
  unverified:
    packet: not_run
    annotation: wdc.challenge_survivability_unverified
    checker_route: HUMAN_GATE
  failed:
    packet: fail
    checker_route: FAIL
  not_applicable:
    packet: not_applicable
```



```

    checker_route: ordinary_SEARG_route
    unknown:
      packet: not_run
      annotation: wdc.relevance_unknown
      checker_route: HOLD
    not_run:
      packet: not_run
      checker_route: HOLD_or_HUMAN_GATEDepending_on_materiality

verification_result:
  wrcg_projection_total: true
  cst_projection_total: true
  unprojectable_values_found: []
  diagnostic_preservation:
    packet_enum_alone_is_injective: false
    sidecar_annotation_checker_route_required: true
    package_control_projection_lossless: true
    fail_closed_for_unprojectable_wrcg: true
    material_cst_not_run_is_not_clean_pass: true
    pass_with_limits_as_output: false
    textual_updates_required:
      "03": false
      "07": false
      "04": false

gate_candidate_state: "SOUND_FOR_REVIEW"
gate_closure_state: "NOT_CLOSED_BY_THIS_RECORD"
recommended_next_artifact: "WDC_ASM_003_WRCG_CST_PROJECTION_VERIFICATION_REVIEW_RECORD_v0_1.md"

```

12. Regression / nonclaim check

This record does not change the package corpus.

It preserves:

```

03 packet field names and enums;
07 WRCG/CST projection crosswalks;
04 canonical checker vocabulary;
08 fixture-output boundary;
WDC-ASM-002 closed state;
WDC-ASM-001/004/005/006 pending state;
release / conformance / deployment / witness-independence nonclaims.

```

No source artifact is edited by this record.

No package assembly is performed.

13. Verification conclusion

WDC-ASM-003 verification candidate: SOUND.

The 07 WRCG and CST sidecar vocabularies project into the accepted 03 packet enums without unprojectable values.

The mapping is not naively lossless at the packet-field level, because richer states such as `pass_with_gates`, `warning`, `hold`, `unverified`, and `unknown` are intentionally compressed.

The mapping is acceptable at package-control level because the missing bits are preserved by:

```

sidecar result/outcome;
canonical wdc.* annotation;
checker route;
evidence / reviewer refs;
materiality and trigger context;
fail-closed behavior for unprojectable states.

```

Therefore this record recommends independent review of this verification candidate. If review returns PASS, c/a may create a separate WDC-ASM-003 closure record and the manifest may be updated to:

```

WDC-ASM-003: CLOSED
projection_verified: true

```

without changing package assembly, release, conformance, deployment, or witness-independence claims.

14. Closing

WDC-ASM-003 / SEARG-WDC-0Q-005 verification prepared.
WRCG projection: total, sidecar-preserved, fail-closed.
CST projection: total, sidecar-preserved, material not_run cannot become clean pass.
No unprojectable values.
No textual updates required to 03/07/04.
No gate closure performed by this record.
Next: WDC_ASM_003_WRCG_CST_PROJECTION_VERIFICATION_REVIEW_RECORD_v0_1.md.